

Intro to Signalling

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Introduction

We now change the timing of events and look at signalling games

- We have seen that asymmetric information about essential economic characteristics leads to efficiency problems when the informed parties' information affects their market behaviour.
- Like adverse selection and screening, signalling is relevant in situations where asymmetric information exists before the contract is offered.
- We allow the informed party to move first and try to “signal” his private information to the other party
 - The signal is something which is relatively more costly for some types than others, or benefits some types more than others.
 - Example: Car seller can offer a warranty

Similarities and differences to the screening model:

- The structure of the signalling model is different than the screening model
 - With **screening**, the **uninformed** parties develop mechanisms to distinguish between informed individuals.
 - With **signalling**, it is the **informed** party that signals to reveal his type.
- In the screening model, beliefs are entirely determined from nature's probability distribution.
 - I.e. with two types we essentially solve two decision trees simultaneously, weighting outcomes with nature's probability distribution.
- With signalling, beliefs (i.e. a player's perception about which node she is at) can be a function of the signals sent (because different types of agents may choose different signals)
 - but the signals are also a result of the initial beliefs which are held by the principal. So this is recursive.
 - Signalling can be constrained inefficient due to self-fulfilling beliefs.

Signalling

- Signalling games are examples of dynamic games with incomplete information
- We now have a signal extraction problem, which is not directly a problem of incentive compatibility
- But incentive compatibility is of course key in solving for the equilibrium (i.e. given beliefs and strategies of P , the agents' actions in eqm need to be incentive compatible)
 - The party with private information will try to use it in his own personal interest. Will send signal if it is in his personal interest to do so.
- Equilibrium concept is Perfect Bayesian Equilibrium (PBE). Beliefs are important and need to be "correct" in equilibrium.

Example: Why do you obtain a college degree? (Spence, 1973)

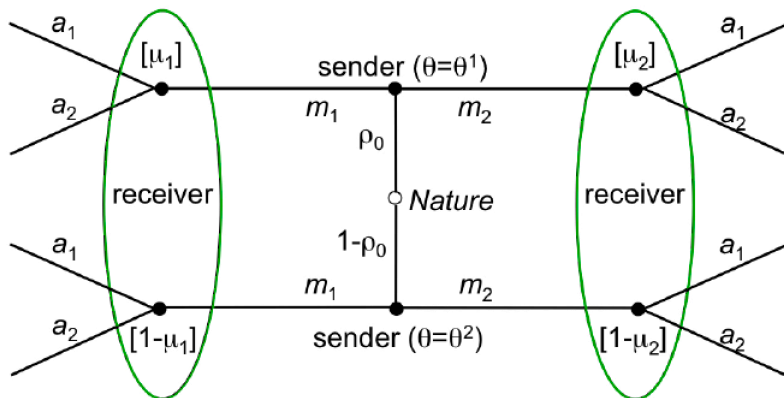
- Maybe because you think that what you learn in the courses will prepare you for your job.
- Other argument: Nothing you learn is useful in your future job. But you need a degree to prove to potential employers that your ability is high. How does a degree prove this point? The cost of obtaining a degree is higher for a person with low ability than for one with high ability, so the low ability person cannot easily imitate the high ability one.
- Importantly, we need the signals to be credible. Quoting Spence: "*...signals, that would carry information persistently in equilibrium from those with more to those with less information.*"

Signalling games - recap

Dynamic game of incomplete information with a Sender (S) and a Receiver (R). The timing is:

- Nature chooses type θ of the sender: $\theta \in T = \{\theta_1, \dots, \theta_I\}$ according to a probability distribution p with $\sum_{i=1}^I p(\theta_i) = 1$.
- The sender (Agent) observes his θ and chooses a message $m \in M = \{m_1, \dots, m_J\}$.
- The receiver (Principal) observes m (and forms a belief of type) and chooses action $a \in A = \{a_1, \dots, a_K\}$.
- Pay-offs are realized ($U_S(\theta, m, a)$ and $U_R(\theta, m, a)$).

Signalling games - recap



Terminologi

- Singleton information sets: I know which specific node I'm at.
- Non-singleton information sets : I don't know which node I'm at but form beliefs.
 - in a separating equilibrium, beliefs at each information set are degenerate, i.e. assign probability 0 or 1 to each node.
 - in a pooling equilibrium, beliefs are determined by nature's probability distribution.
- Strategy: A complete contingent plan specifying what the player will do at each information set where he is potentially called to move.
- The equilibrium path: The path we take through the game tree when each player plays the actions that are prescribed by the equilibrium strategies.
- Being off the equilibrium path: Being in parts of the game tree that are not reached by the actions that are prescribed by the equilibrium strategies.

Signalling games - recap

Two types of equilibria: pooling and separating

- Due to incomplete information beliefs are a crucial feature
- Beliefs are in some sense also a “problem” here in terms of deriving predictions which are unique and “credible” ...
 - Prediction of labor market model under competitive screening: unique equilibrium - if one exists
 - With signalling: multiple equilibria - both pooling and separating.
- Question: In which type of equilibrium would signalling allow us to circumvent problems of adverse selection?

Different concepts

- NE: Nash Equilibrium. In normal form static games (and also in other games).
- SPNE: Subgame Perfect Nash Equilibrium: In extensive form games.
- BNE: Bayesian Nash Equilibrium. In normal form static games of incomplete information.
- PBE: Perfect Bayesian Equilibrium: In extensive form games of incomplete information.
 - Replace the consistency condition in the definition of SPNE with restrictions on how beliefs are formed and updated:
 - If an information set " I " is reached with positive probability under the equilibrium strategy profile σ (is on the equilibrium path), then the belief at " I " is derived from σ using Bayes' rule.

Compare SPNE and signalling game

- In games of incomplete information, some information set is not a singleton.
- A subgame must start at singleton information sets, so SPNE can not be used to refine prediction.
- Replace idea of subgame with the more general idea of a continuation game
- Can start at a non-singleton information set.
- At each information set, the player who moves has belief about which node of the information set he is at.
- These beliefs are required to be updated in accordance with Bayes rule whenever applicable. [ensures that beliefs are “right” in eqm]
- Idea of perfect Bayesian equilibrium: combines the ideas of subgame perfection, Bayesian equilibrium and Bayesian inference (“optimal processing of information”)

Perfect Bayesian EQ

Perfect Bayesian equilibrium is a strategy profile σ^* and a system of beliefs μ^* satisfying Requirements 1-3 below.

Requirement 1: At each information set, the player who moves must have a belief about which node in the information set has been reached by the play of the game. For a non-singleton information set, a belief is a probability distribution over the nodes in the information set; for a singleton set, the player's belief puts probability one on the single decision node.

Perfect Bayesian EQ

Requirement 2: Given their beliefs, the players' strategies must be sequentially rational. That is, at each information set the action taken by the player with the move (and the player's subsequent strategy) must be optimal given the player's belief at the information set and the other player's subsequent strategy (where a subsequent strategy is a complete plan of action covering every contingency that might arise after the given information set has been reached).

Requirement 3: The players' beliefs in every information set (on and off the equilibrium path) are determined by the players' equilibrium strategies and Bayes rule whenever applicable.

Signalling in labour market

We look at the labour market example discussed in previous lectures (now modified to a setting with signalling)

- An individual has more information about his abilities and past experiences than the firm that he is applying to.
- All individuals would like to pass themselves off as attractive employees.
- The basic idea: more productive individuals can attempt to differentiate themselves from the less productive ones by investing in a costly signal.
 - To be credible this signal must be too costly for the less productive individuals to imitate.
- One possible signal is the education level of a worker: higher education level is easier to acquire for more productive individuals.

Game

- There is a single worker, who can be of two types, θ_L or θ_H , with $\theta_H > \theta_L \geq 0$, and two firms.
- The worker can acquire education e , which is observable, but not directly productive.
- The worker has outside option $r(\theta_H) = r(\theta_L) = 0$.
- A worker with productivity θ who accepts a contract with wage w and education level e has utility

$$U(w, e, \theta) = w - c(e, \theta).$$

Single crossing assumptions

We assume that

$$c(0, \theta) = 0 \quad \forall \theta \in \{\theta_L, \theta_H\}$$

$$c_e(e, \theta) > 0 \quad \forall e > 0, \forall \theta \in \{\theta_L, \theta_H\}$$

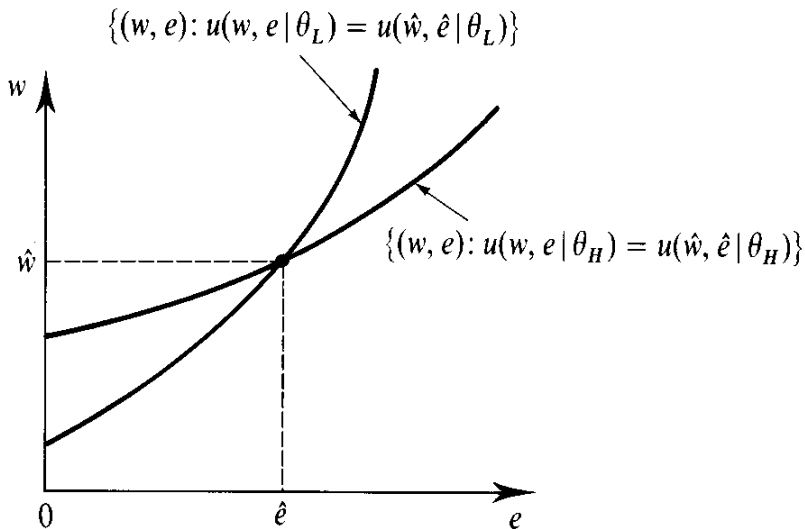
$$c_{ee}(e, \theta) > 0 \quad \forall \theta \in \{\theta_L, \theta_H\}$$

$$c(e, \theta_L) > c(e, \theta_H) \quad \forall e > 0$$

$$c_e(e, \theta_L) > c_e(e, \theta_H) \quad \forall e > 0$$

where $c_e(e, \theta) = \frac{\partial c(e, \theta)}{\partial e}$.

- Worker's indifference curve:
 $u(w, e, \theta) = \bar{u} \Leftrightarrow w - c(e, \theta) = \bar{u}.$
- MRS between wage and education at (w, e) is $\frac{dw}{de} = c_e(e, \theta).$
- Marginal cost of education is more costly for the low-ability type.
- Low type has steeper indifference curves in (e, w) -space.
- Hence we have the single-crossing property.



Firms

- There are two firms, 1 and 2.
- If either firm employs the worker, and the worker's type is θ , it obtains a profit

$$\Pi(\theta, w) = \theta - w$$

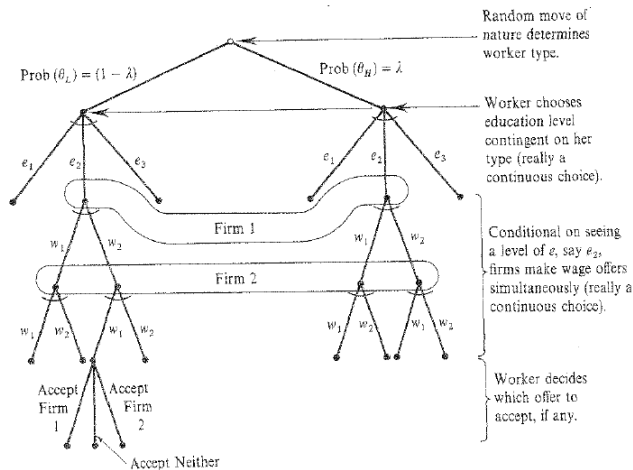
when it pays wage w .

- Note that we assume that education does not affect productivity at all.
- Firms compete à la Bertrand.

Game

Timing:

- 1 Nature draws the worker's ability (type) $\theta \in \{\theta_L, \theta_H\}$. The prior probability of θ_H is λ .
- 2 The worker learns his θ and chooses an education level $e \in \mathbb{R}_+$. That is, the worker is the sender, and the education level is the message.
- 3 The firms observe e , but not θ . The firms simultaneously make wage offers to the worker. That is, the firms are the receivers, and the wage offers are the actions.
- 4 The worker accepts the highest wage offer. If he is indifferent, he randomizes 50-50. Firms are in Bertrand competition.



Strategies

- The worker's strategy: $e(\theta)$ and which offers to accept.
- Firm i 's strategy: $w_i(e)$, $i=1,2$.

First best solution

- First best solution is as discussed in previous lectures:

$$w(\theta_i) = \theta_i$$

$$e_i = 0$$

for $i = 1, 2$.

Second best analysis

Now assume θ is unobservable to the firm. Education is available as specified above.

- Denote by $\mu(e)$ the firms' beliefs about the worker type, i.e. $\mu(e)$ is the prob firms assign to worker being high type given the worker has education level e .
- So $\mu(e) \in [0, 1]$ represents the belief (subjective probability) of meeting a θ_H -worker after observing education level e .
- Solving for the equilibrium we specify what $\mu(e)$ looks like.

Perfect Bayesian Nash Equilibrium

A set of strategies $\{e(\theta), \{w_i(e)\}_{i=1}^2\}$ and a belief function $\mu(e) \in [0, 1]$ is a PBE if:

- 1 The worker's strategy is optimal given the firms strategies (including beliefs).
- 2 The belief function $\mu(e)$ is derived from the worker's strategy using Bayes rule where possible.

Furthermore, in this context:

- 3 The wage offers following each choice e constitute a Nash Equilibrium of the simultaneous-move wage offer game in which the probability that the worker is of high ability is $\mu(e)$.

Note:

- "Where possible" in (2) means that off the eqm path we can place "arbitrary" beliefs.
- $0 \leq \mu(e) \leq 1$ (otherwise it is not a belief).
- Strictly speaking the worker's strategy also includes his accept/reject decision of the wage offer. Firms' competition makes the wage offers attractive.

Solve for equilibria

Solve by working backwards.

- Condition (3) implies that both firms, $i = 1, 2$, will offer the wage

$$w_i(e) = \mu(e)\theta_H + (1 - \mu(e))\theta_L \equiv w(e)$$

- implying that $\theta_L \leq w(e) \leq \theta_H$.
- I.e. wage equals the expected marginal product.
- Remember firms are in Bertrand competition.
- The wage function $w_i(e)$ is firm i 's strategy.